

DEXTER FERNANDES

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SUMMARY

Computer Vision Engineer with 3+ years building real-time vision and SLAM systems in PyTorch and C++17, spanning neural network training through embedded deployment. Built a LiDAR SLAM system with visual odometry on a Clearpath Husky, running the full perception stack on an NVIDIA Jetson AGX Xavier SOM. Shipped asynchronous C++17 inference pipelines on Intel Movidius VPUs, doubling throughput under hard real-time constraints. MSc Robotics (Merit), University of Bristol.

RELEVANT EXPERIENCE

Esbaar — AI Engineer

01/2023 – 08/2023 · MUSCAT, OMAN

- Owned the end-to-end ML lifecycle for real-time perception on BP rigs, from data curation and training through deployment and on-site validation, improving F1 by 15% under 24/7 operation.
- Adapted YOLOv5 detectors in PyTorch, replacing backbone and neck layers and retraining for rig-specific classes, with Optuna sweeps and Weights & Biases tracking across runs.
- Hardened deployed services with Dockerised microservices, Kafka messaging, health checks and safe restarts, cutting live video pipeline crashes by 67% and enabling recovery without site visits.
- Applied Bayesian optimisation, ablation studies, hypothesis testing and confidence intervals in Jupyter to support model selection, threshold tuning and failure analysis.
- Ran TIDE error analysis on COCO-format production data to split live detector failures by error type, directing dataset fixes and threshold tuning at the dominant one.
- Developed NVIDIA DeepStream 6.x and FFmpeg video pipelines for 24/7 perception on BP drilling rigs, using multiprocessing to isolate capture from inference so one stalled stream couldn't block the rest.

LivNSense Technologies — Lead Software Engineer, AI

05/2020 – 12/2022 · BENGALURU, INDIA

- Built asynchronous C++17 perception pipelines on OpenVINO across Intel Movidius VPUs and Core i7 industrial edge boxes, overlapping preprocessing, inference and post-processing to raise throughput from 5 to 15 FPS under real-time constraints.
- Implemented optical flow based ego-motion estimation (visual odometry) feeding PyTorch CNN classifiers to infer vehicle state, supporting collision avoidance on a moving construction vehicle.
- Profiled C++ inference hotspots with Intel VTune and integrated Boost C++ libraries into production pipelines, then wrote gtest unit tests and documented interfaces, catching regressions before deployment.
- Built a real-time multi-camera perception pipeline on a moving construction vehicle, synchronising streams and running incident detection under edge compute constraints.
- Optimised architectures with quantisation-aware training, structured pruning and layer fusion in Keras and OpenVINO, cutting model size by 50% while holding production accuracy on Intel edge targets.
- Led four engineers building a two-stage detector feeding helmet-colour classification for construction safety, cutting false positives by 33% within latency and power budgets.

PROJECTS

DeepStream RTSP Pipeline github.com/Dexter-Fernandes/deepstream-rtsp-pipeline

06/2026

- Built an end-to-end DeepStream 9.0 and GStreamer RTSP pipeline running YOLO object detection and NvDCF tracking across 3 concurrent feeds, with metadata extraction and re-streamed output.
- Converted YOLO26n to a dynamic-batch TensorRT FP16 engine on 6GB VRAM Turing hardware, validating FP16 against FP32 accuracy and profiling p99 latency to sustain 29.7 FPS per stream live.
- Implemented a C++ and CUDA TensorRT decode plugin with IPluginV2DynamicExt, moving post-processing into the inference engine to support custom-layer deployment.
- Benchmarked IoU, NvDCF and NvSORT trackers on MOTA, IDF1, ID switches, fragmentation, FPS and VRAM impact to guide tracker selection.

Modern Edge CV Benchmarking github.com/Dexter-Fernandes/ModernEdge-CV-Benchmarking

03/2026

- Benchmarked modern PyTorch detectors including YOLOv8, YOLOv10, YOLO26 and RT-DETR on accuracy, latency and model size to compare deployment suitability for edge targets.
- Extended the comparison to EfficientViT, MobileSAM and Depth Anything v2 to assess whether segmentation and depth models are viable under the same edge constraints.
- Exported models to ONNX and tested OpenVINO deployment paths, comparing mAP, inference time and model size to support model selection for low-latency systems.
- Evaluated Vision Transformer and foundation-model approaches across detection, segmentation, depth estimation and open-vocabulary understanding.

ROS2 SLAM and Nav2 Simulation

02/2026

- Built a simulation-based robotics stack in ROS2 Jazzy and Gazebo Harmonic to evaluate LiDAR-centric SLAM, LiDAR and RGB fusion SLAM, and autonomous navigation in structured scenarios.
- Benchmarked RTAB-Map, LIO-SAM, Cartographer and GLIM across 2D and 3D LiDAR configurations, comparing mapping and localisation behaviour to support system selection.

- Developed ROS2 nodes, launch files and parameterised configs in C++ and Python, using tf2, message synchronisation and timestamp alignment to debug transform timing issues.
- Implemented point cloud filtering, segmentation and registration with PCL and Open3D, and added an EKF localisation layer with robot_localization fusing wheel odometry and IMU.

EDUCATION

University of Bristol — MSc Robotics, Merit09/2023 – 09/2024 · BRISTOL, ENGLAND

Dissertation: Radiological SLAM with LiDAR Odometry

- Built a real-time SLAM system in C++17 and ROS2 (roslcpp) on a Clearpath Husky, fusing Velodyne LiDAR, RealSense depth and radiological sensing for inspection mapping.
- Ran the full SLAM and perception stack on an NVIDIA Jetson AGX Xavier SOM under JetPack, working within the platform's compute and memory limits for on-robot operation.
- Configured Ceres, g2o and GTSAM as solver backends and tuned ICP parameters across environments.
- Configured Nav2 in simulation for autonomous waypoint traversal over the generated map, testing survey routes before running them on the Husky.

Visvesvaraya National Institute of Technology — BTech Electrical and Electronics Engineering07/2016 – 05/2020 · NAGPUR, INDIA

Foundation in signals, systems, embedded hardware and control, which later supported work in real-time AI, sensor systems and robotics software.

SKILLS

Languages	Python, C++17, CUDA C++, Bash
CV and Deep Learning	PyTorch, TensorFlow, Keras, OpenCV, scikit-image, CNNs, object detection, image classification, semantic and instance segmentation, tracking, optical flow, data augmentation, YOLO (v5/v8/v10/v26), RT-DETR
Robotics and SLAM	ROS2 (Robot Operating System 2), roslcpp, tf2, rosbag2, Gazebo, RTAB-Map, LIO-SAM, Cartographer, GLIM, Nav2, LiDAR SLAM, visual odometry, ICP (Iterative Closest Point), pose graph optimisation, scan matching, PCL, Open3D, Eigen, Ceres, g2o, GTSAM, robot_localization, EKF
Edge Inference	TensorRT (custom plugins, dynamic batching), OpenVINO, ONNX, quantisation-aware training, FP16/INT8, structured pruning, layer fusion, latency and throughput profiling, NVIDIA Jetson AGX Xavier (SOM), Intel Movidius VPU, Intel VTune
Applied Mathematics	linear algebra, probability, Bayesian optimisation, hypothesis testing, ablation studies, confidence intervals
Video Pipelines	NVIDIA DeepStream, GStreamer, FFmpeg, RTSP, nvinfer, nvtracker, multi-camera synchronisation, ISP and camera tuning (RGB and IR)
Experimentation and MLOps	Optuna, Weights & Biases, Jupyter, DVC (Data Version Control), CVAT, benchmarking harnesses, TIDE (Toolbox for Identifying Detection Errors)
Production and Tooling	Docker, CI/CD, GitHub Actions, Git, CMake, pytest, unittest, gtest, Boost, multiprocessing, REST API design, Linux
Professional	technical leadership, mentoring, Agile and Scrum, cross-functional delivery

CERTIFICATIONS AND AWARDS

Deep Learning Specialisation, deeplearning.ai (07/2019)

Best Speaker award, ABC Toastmasters Club (01/2020)